

Introduction to Computer Architecture

Sheet #1

Questions:

1. What is the difference between computer architecture and computer organization?
2. Name and explain the main components of a Von Neuman computer.
3. Von Neuman was responsible for several major contributions to computer design.
Explain.
4. Describe the Fetch-Decode-Execute instruction cycle.
5. What are the three main components of a computer?
6. What's the difference between Cache and RAM?
7. State the computer levels from high to low.
8. What's the difference between SRAM and DRAM?
9. What is ROM? And what's its types?
10. If a ROM has a total capacity of 16K bits and 2 outputs, how many address lines are there?
11. What is the Von Neumann bottleneck and how does it impact system performance?
12. The Von Neuman architecture includes all of the following except:
 - a. A stored program
 - b. Sequential processing of instructions
 - c. Parallel processing
 - d. A CPU, memory, and I/O system
 - e. The Von Neuman architecture contains all of these components
13. Does Ram differ from ROM? If yes, how so?
14. What are non-von Neumann models, and how do they differ from the traditional von Neumann architecture?